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Right Question is Already Half the Answer: Fully Unsupervised LLM Reasoning Incentivization

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Abstract

While large language models (LLMs) have demonstrated exceptional capabilities in challenging tasks such as mathematical reasoning, existing methods to enhance reasoning ability predominantly rely on supervised fine-tuning (SFT) followed by reinforcement learning (RL) on reasoning-specific data after pre-training. However, these approaches critically depend on external supervisions–such as human labelled reasoning traces, verified golden answers, or pre-trained reward models–which limits scalability and practical applicability. In this work, we propose Entropy Minimized Policy Optimization (`EMPO`), which makes an early attempt at fully unsupervised LLM reasoning incentivization. `EMPO` does not require any supervised information for incentivizing reasoning capabilities (i.e., neither verifiable reasoning traces, problems with golden answers, nor additional pre-trained reward models). By continuously minimizing the predictive entropy of LLMs on unlabeled user queries in a latent semantic space, `EMPO` enables purely self-supervised evolution of reasoning capabilities with strong flexibility and practicality. Our experiments demonstrate competitive performance of `EMPO` on both mathematical reasoning and free-form natural reasoning tasks. Specifically, without any supervised signals, `EMPO` boosts the accuracy of Qwen2.5-Math-7B Base from 30.7% to 48.1% on mathematical benchmarks and improves the accuracy of Qwen2.5-7B Base from 32.1% to 50.1% on MMLU-Pro.

1 Introduction

Large language models (LLMs) have demonstrated exceptional potential in challenging tasks such as mathematical reasoning [1] and code generation [2]. A prevailing paradigm for training reasoning LLMs involves firstly performing supervised fine-tuning (SFT) and then reinforcement learning (RL), or iterative combinations of both, applied to reasoning-specific datasets after pretraining [3]. Unfortunately, these methods typically depend on large-scale reasoning datasets with various forms of supervised information, such as human-labeled reasoning traces, verified golden answers, or an additional pre-trained reward model. As a consequence, endowing LLMs with powerful reasoning capability through human experts is becoming increasingly time-consuming and costly, which greatly limits the scalability and broader adoption of reasoning models.

To mitigate this, previous work employs self-consistency to construct pseudo data and deploy supervised finetuning for better performance [4]. However, performance improvement is limited and under risks of model collapse [5]. Recent advancements, such as Jiao et al. [6], frame the labeling of solutions as evaluation against test cases. However, the proposed method still necessitates supervision from instruction finetuning data and supervision signals from the frontier LLMs to initialize the RL process. Another more recent work [7] introduces a two-stage framework to construct self-rewarding

Work in progress.

reasoning models using self-generated data followed by RL. Despite the superior performance, the proposed method relies on a ground-truth verifier to obtain self-correction reasoning traces by rejection sampling. These approaches inspire our exploration of a critical open question: **How can we incentivize LLM reasoning capacities in a fully unsupervised manner?**

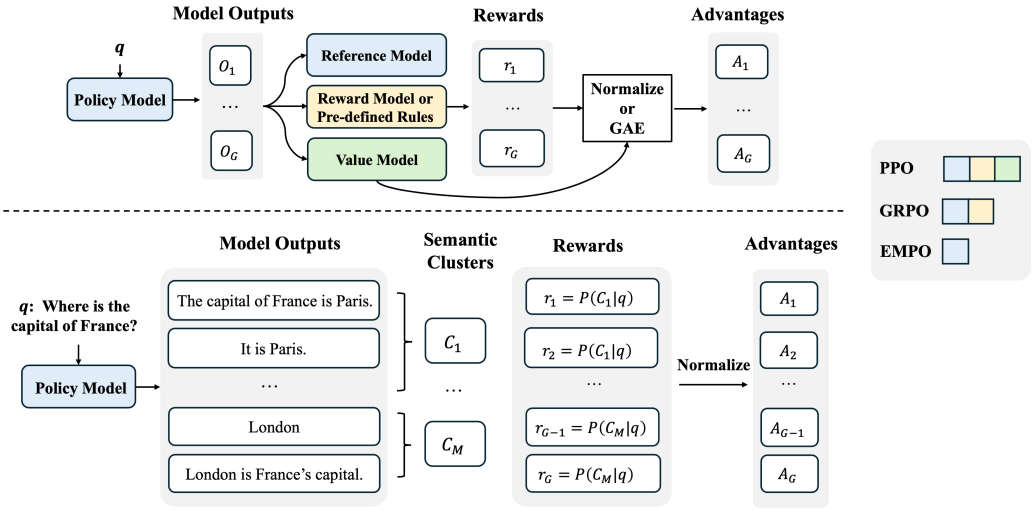


Figure 1: Overview of the proposed method. Previous method like PPO [8] or GRPO [9] typically relies on external supervised signals, e.g., a pretrained reward model or golden answers. The proposed Entropy Minimized Policy Optimization (EMPO) samples a set of responses from the current policy model, and then builds semantic clusters according to their equivalence. By continuously minimizing the entropy at a meaning level, our method achieves competitive benchmark performance without any external supervision, i.e., rule-based reward, pre-defined test cases or an pre-trained reward model.

Recent advanced DeepSeek-R1-Zero [10] demonstrates robust reasoning capabilities without dependency on SFT data. By directly initiating RL from the base model, DeepSeek-R1-Zero autonomously evolves sophisticated reasoning behaviors such as reflection and self-critic by exploring the reward signals provided by rule-based rewards. i.e., verified golden answers or an additional pre-trained reward model. Inspired by the success of DeepSeek-R1-Zero, our motivation is to devise a fully unsupervised approach for powerful reasoning capability. Specifically, we propose a novel reinforcement learning algorithm termed as Entropy Minimized Policy Optimization (EMPO), which incentivizes the reasoning capability of LLMs in a fully unsupervised manner by minimizing their predictive entropy in a latent semantic space. This method optimizes the model to favor reasoning traces yielding consistent answers, enhancing output reliability. The semantic entropy objective we propose to minimize is a well-established measurement of LLMs' uncertainty, which extends beyond mathematical reasoning to free-form question-answering tasks. We further introduce entropy thresholding to filter unreliable reasoning traces, stabilizing the unsupervised training process. Experiments on various tasks including mathematical reasoning and free-form question-answering are conducted to validate the proposed method. Our contributions are summarized as follows:

- We propose an effective and principled strategy called Entropy-Minimized Policy Optimization (EMPO) for incentivizing the reasoning capabilities of LLMs in a fully unsupervised manner.
- We provide primary analysis with insights that semantic entropy has strong negative correlation with model accuracy, and thus it can be viewed as a promising proxy objective in the fully unsupervised setting. Besides, we leverage a simple entropy thresholding to avoid potential reward hacking.
- Experiments on both mathematical reasoning tasks with deterministic golden answers and free-form natural reasoning tasks are conducted to validate the efficacy and versatility of EMPO.

2 Related Work

Self-Supervised and Semi-Supervised Reasoning. To address the dependency on labeled data, several self-supervised and unsupervised methods have emerged. Huang et al. [4] propose a self-improvement framework where LLMs generate high-confidence answers using Chain-of-Thought (CoT) prompting and self-consistency, subsequently fine-tuning on these pseudo-labels. However, the performance gains are often limited, and there is a risk of model collapse, as noted in [5]. Recently, Patel et al. [11] apply self-improvement to web navigation tasks in WebArena, fine-tuning on synthetic data generated by the model itself. Li et al. [12] enhance long-context reasoning via SeaLong, sampling multiple outputs and optimizing with Minimum Bayes Risk. These methods, while reducing reliance on external labels, still involve supervised fine-tuning steps, contrasting with EMPO's fully unsupervised RL approach.

Self-Rewarding and RL-based Reasoning. RL has become a prominent technique for enhancing LLM reasoning, often leveraging external or self-generated rewards. Yuan et al. [13] propose using the LLM itself via LLM-as-a-Judge prompting to provide rewards during training, reducing reliance on human feedback. Similarly, Xiong et al. [7] propose a two-stage self-rewarding framework for mathematical reasoning, generating data and applying RL with a ground-truth verifier for self-correction, achieving superior performance but requiring supervised signals. Jiao et al. [6] frame solution labeling as evaluation against test cases, yet still rely on instruction fine-tuning and frontier LLM signals for RL initialization. Wen et al. [14] introduce Entropy-Regularized Token-Level Policy Optimization (ETPO), augmenting RL with an entropy bonus to promote exploration, differing from EMPO's entropy minimization focus. Guo et al. [10] with DeepSeek-R1 demonstrate robust reasoning via RL from a base model, using rule-based rewards. Xi et al. [15] present R^3, a reverse curriculum RL approach using outcome supervision to mimic process supervision benefits. Wang et al. [16] propose CREAM, which enforces consistency regularization between internal reward models during self-training. These methods highlight a spectrum of supervision levels, positioning EMPO as unique in its fully unsupervised nature, leveraging semantic entropy as an internal reward.

Entropy Minimization and Semantic Consistency. Entropy minimization is a well-established technique in semi-supervised and unsupervised learning, with roots in traditional machine learning. Grandvalet and Bengio [17] demonstrate that minimizing entropy on unlabeled data can improve classification accuracy by encouraging model confidence. Test-time adaptation methods like Tent [18] adapt models to new domains by minimizing entropy on test data, filling domain gaps without additional labels. More recent work, COME, [19] extends this principle to conservative entropy minimization for robust adaptation. These approaches highlight the potential of entropy minimization as an unsupervised objective, which EMPO leverages for LLM reasoning by extending it to semantic entropy [20] in a latent space. Farquhar et al. [21] further validate semantic entropy's utility in detecting hallucinations, reinforcing its relevance. Kharitonov et al. [22] explore entropy minimization in emergent languages, finding it naturally aligns with successful communication, providing additional theoretical foundation for EMPO.

3 Method

We propose an RL-based method to minimize the entropy of LLM generations in a latent semantic space for incentivizing its reasoning capability. We term our method Entropy-Minimized Policy Optimization (EMPO), which is devised in a fully unsupervised manner without any forms of supervised information.

3.1 Preliminaries

Recent advancements in reinforcement learning have demonstrated remarkable breakthroughs in enhancing the reasoning capabilities of LLMs. Taking the representative RL technique Group Relative Policy Optimization (GRPO) [9] used by DeepSeek-R1-Zero [10] as an example. GRPO first samples a group of outputs $\{o_1, \cdots, o_G\}$ from the policy model π_θ and then optimizes it by maximizing the

following objective:

$$\mathcal{J}_{\text{GRPO}} = \mathbb{E}_{[q\sim P(Q),\{o_i\}_{i=1}\sim\pi_{\theta(O|q)}]} \left[\frac{1}{G}\sum_{i=1}^{G}(\min(A_i, \text{clip}(1, 1-\epsilon, 1+\epsilon)A_i) - \beta\, KL(\pi_\theta||\pi_{ref}) \right], \tag{1}$$

where β is a hyper-parameter which avoids the policy model to diverge too far away from the reference model π_{ref}. ϵ clips extreme advantages for stability. G is the number of samples in one group. A_i is the advantage computed by normalizing the rewards within each group as $A_i = \frac{r_i - mean(\{r_1,\cdots,r_G\})}{std(r_1,\cdots,r_G)}$. In math reasoning task, the reward can be computed by rules:

$$r_i = \begin{cases} 1 & \text{if verifier}(o_i, a) = \text{True} \\ 0 & \text{otherwise} \end{cases}, \tag{2}$$

where a verifier is used to determine the correctness of o_i by comparing it with the golden answer a.

Unlike the above example, we consider fully unsupervised optimization settings where there are no golden answers to verify the correctness of model predictions. In this circumstance, we only have unlabeled reasoning problems $P(Q)$. Such problems were freely raised by users during the deployment of LLMs. Given a pre-training LLM π_θ parameterized by θ, our goal is to enhance its reasoning ability by only utilizing the unlabeled user problems $\{q_i\}_{i=1}^n$, which requests minimized cost of data collection.

3.2 Semantic Entropy Minimization Objective

Entropy is a classical unsupervised objective in the traditional semi-supervised and unsupervised learning fields [17, 23]. Previous works in computer vision show that by continuously minimizing the entropy on unlabeled samples after pre-training, the classification accuracy of machine learning models can be significantly improved to fill the domain gaps [18, 19]. The basic intuition behind entropy minimization is that a robust model should not only fit labeled data well but also make confident and consistent predictions on unlabeled data. This principle encourages the model to avoid ambiguity and make decisive predictions, thereby enhances generalization. In this work, we choose semantic entropy [20] as our unsupervised optimization objective, which is a natural extension of classical Shannon entropy specified for large language models. Intuitively speaking, minimizing semantic entropy encourages the LLMs' outputs to be more consistent in semantic level rather than format, and thus the final answers are expected to be more reliable.

Specifically, semantic entropy first samples a group of outputs $\{o_1, \cdots, o_G\}$ and then clusters the output sequences according to their meaning. That is, if two outputs share the same meaning (i.e., they are bidirectionally entailed), they should be merged into one same cluster in the semantic space. As we will show later, this can be done without notable computational cost by predefined rules such as N-gram, regular expressions or an additional (bert-like) small language model. Once built such a set of meaning clusters $\{c\}$ in semantic space, we then approximate the probability over the meanings as the proportion of sampled answers as

$$p(c_j|x) \approx |c_j|/G, \tag{3}$$

where $c_j \in \{c\}$ is the j-th meaning cluster. $|c_j|$ denotes the numbers of outputs that belong to c_j. Finally, given question q, the semantic entropy (denoted as H) over the model's output meanings distribution can be estimated as follows

$$H = -\sum_{c_j\in\{c\}} p(c_j|q)\log p(c_j|q). \tag{4}$$

As proven by previous work, semantic entropy has a strong negative relationship with model accuracy, which can be used as an efficient measurement to detect unreliable LLM generations such as confabulation and hallucination [20, 21]. Motivated by this, we propose to leverage semantic entropy as an unsupervised optimization objective for incentivizing the reasoning capability of LLM.

3.3 Entropy-Minimized Policy Optimization

We propose Entropy-Minimized Policy Optimization (`EMPO`), an RL-based method that optimizes the pre-trained large language model π_θ to favor low semantic entropy responses given unlabeled

user questions $\{q_i\}_{i=1}^{n}$. Given input questions, EMPO incentivizes the outputs that belong to higher probability meaning cluster, and thus minimizes the semantic entropy over the meaning distribution. Specifically, given a question q, our EMPO first samples a group of output $\{o_1, \ldots, o_G\}$ from the current model π_θ and then merges them into a set of M meaning clusters $\{c_1, \ldots c_M\}$. Once built such a meaning set, EMPO approximately minimizes the semantic entropy H by maximizing the following objective

$$\mathcal{J}_{\texttt{EMPO}} = \mathbb{E}_{[\{q\}\sim P(Q),\{o_i\}_{i=1}^{G}\sim\pi_\theta(O|q)]} \frac{1}{|G|}\sum_{i=1}^{|G|}(A_i),\ A_i = \frac{r_i - mean(\{r_1,\cdots,r_G\})}{std(r_1,\cdots,r_G)} \tag{5}$$

where A_i is the advantage of output o_i calculated by normalizing the rewards. Unlike GRPO in which the rewards is calculated depending on external supervision such as pre-defined rules or an reward model, in EMPO, the reward assigned for the i-th outputs o_i is the likelihood of its meaning cluster, i.e.,

$$r_i = p(c_j|q),\ \text{where } l(o_i) = c_j, \tag{6}$$

where the meaning likelihood $p(c_j|q)$ is approximated by Eq. 3. Intuitively, the outputs convey higher-probability meanings are of higher advantages, and are therefore incentivized through training. To further stabilize the optimization, we introduce entropy thresholdings to filter out user queries q that result in overly high or low entropy unreliable answers. The final optimization objective of EMPO is

$$\begin{aligned}\mathcal{J}_{\texttt{EMPO}} = \mathbb{E}_{[\{q\}\sim P(Q),\{o_i\}_{i=1}^{G}\sim\pi_\theta(O|q)]} \\ \left[\frac{1}{|G|}\sum_{i=1}^{|G|}(\min(A_i, \text{clip}(1,1-\epsilon,1+\epsilon)A_i) - \beta\, KL(\pi_\theta||\pi_{\text{ref}})\right], \\ \text{s.t. } \delta_{low} < H < \delta_{high}\end{aligned} \tag{7}$$

where H is the semantic entropy defined in Eq. 4. The questions results in highly unreliable answers with entropy greater than δ_{high} are filtered out. Besides, we also filter out low-entropy answers to maintain the diversity of model outputs and further avoid potential reward hacking. Same with GRPO, an hyper-parameter β controls the strength of KL penalization, which constrains the policy model to not diverge too far away from the reference model. ϵ clips extremely high or low advantages for stability.

4 Experiments

4.1 Experimental Settings

We conduct experiments on multiple datasets including both closed-form math reasoning tasks and free-form natural reasoning tasks. Our EMPO shows competitive performance by purely RL in a fully unsupervised manner compared to supervised finetuning and RL methods.

Prompt Collection. For mathematical reasoning, following the common practice [24, 6, 25], we adopt 20,000 prompts randomly selected from NuminaMath-CoT dataset [26] for training[1]. For free-form natural reasoning tasks, we adopt NaturalReasoning[2], a large-scale dataset consisting of diverse reasoning questions from multiple domains (e.g., Physics, Computer Science, Economics, Social Sciences and more). For training efficiency, we filter out the questions with over-long prompt or reference answer, which results in a subset consists of 80,000 questions.

Evaluation. ∘ For mathematical reasoning, the performance is evaluated on a diverse suite of benchmarks including Minerva Math, MATH, AMC23, OlympaidBench and AIME24. The evaluation codebase is directly borrowed from the SimpleRL project [27], which is consistent with other concurrent works [25]. ∘ For free-form natural reasoning, we evaluate the models on the MMLU-Pro. We prompt the model to reason step by step and output a brief answer in "\boxed{}" and report the multi-choice accuracy. Without specific clarification, all evaluations are conducted using zero-shot prompting and greedy-decoding.

[1] https://huggingface.co/datasets/RLHFlow/numia_prompt_dpo1

[2] https://huggingface.co/datasets/facebook/natural_reasoning

Model training. ◦ For mathematical reasoning tasks, we train Qwen2.5-Math-1.5B and 7B Base models with our EMPO. The baselines we consider include supervised finetuning (SFT), online direct preference optimization (ODPO) [25] and the representative GRPO. We also compared with Qwen2.5-Math Instruction models for a more comprehensive comparison, where the instruction model is trained by iteratively supervised finetuning and RL on private data. ◦ For free-form natural reasoning tasks, we initialize from Qwen2.5-3B and 7B Base models. Different from mathematical reasoning, it is difficult to adopt rule-based reward for free-form question-answering tasks without deterministic golden answers. We consider the corresponding Instruct model, the original Base model with or without few-shot CoT prompt as baselines. Besides, we also compare with SFT where the Base model is tuned to fit the response of Llama3.3-70B-Instruct.

- SFT: We train models by supervised finetuning via Open-Instruct [28] with a learning rate of 1×10^{-6}, a global batch size of 128 and train for 1 epoch with a max length of 2048.
- GRPO: We implement GRPO via trl [29] based on Open-R1 [24]. We sample 7 responses for each prompt ($G = 7$) and adopt the rule-based reward for math reasoning tasks. We use a constant learning rate of 3×10^{-6}, a global batch size of 128 and train the model for 2 epochs with a maximum generation length of 2048. We only use the rule-based accuracy reward and do not adopt format-reward following [30]. The accuracy reward is implemented as follows: If the response contains the correct final answer within "\boxed{}", it receives a reward of 1, otherwise 0. As suggested by recent work [31], we remove the KL penalization term in mathematical reasoning tasks for better performance as well as training efficiency. In free-form question answering, we observed model collapse when training without KL constraint. Thus we adopt $\beta = 0.02$ for KL penalization in all trials.
- Online-DPO: Recent advanced Online-DPO first samples a set of responses and then verifies and selects the responses with highest reward and lowest reward as a preference pair. We directly copy the results from [25], where the model is trained for 7 iterations. Each iteration involves 20K training samples.
- EMPO: EMPO is also implemented via trl based on Open-R1. All the hyper-parameters e.g., number of generations, max generation length, global batch size and KL penalization strength β are the same with GRPO. In mathematical reasoning tasks, we use a set of regular expressions from the commonly-used Math-Verify package to merge the outputs into meaning clusters. For more general free-form natural reasoning, we leverage General-Verifier[3] (a small language model with 1.5B parameters) to determine whether two outputs are of the same meaning or not [20, 21]. Specifically, if the final predictions (i.e., the contents within "\boxed{}") of two model outputs are bidirectionally implicating, then we merge them into one semantic cluster ignoring their reasoning traces.

4.2 Main Results

4.2.1 Performance on Mathematical Reasoning Tasks.

We conduct experiments on mathematical tasks to evaluate our method. The main results are shown in Table 1. EMPO has successfully incentivized the Qwen2.5-Math Base model with reasoning capability without dependency on any external supervision. We observe a substantial improvement in the average performance on commonly used mathematical reasoning benchmarks from 28.1% to 42.1% and 30.7% to 48.1% on 1.5B and 7B models respectively. Notably, through fully unsupervised RL training, the 1.5B and 7B model has both achieved competitive performance (42.1% and 48.1%) near to Qwen2.5-Math-Instruct (40.5% and 48.1%), where the latter depends on carefully-designed data collection and training pipeline (including multi-stage supervised fine-tuning and reinforcement learning).

4.2.2 Performance on Natural Free-form Reasoning Tasks.

We present the results on free-form natural reasoning tasks in Table 2. On the MMLU-Pro benchmark, our EMPO improves the truthfulness accuracy of outputs from 81.96% to 90.83% and 87.16% to 97.25% on Qwen2.5-3B and 7B Instruct model respectively. Besides, on larger-scale trivialQA benchmark, EMPO results in increasing exactly match metric from 45.42% to 58.20% on 3B model,

[3] https://huggingface.co/TIGER-Lab/general-verifier

Table 1: Accuracy on mathematical reasoning benchmarks. We report the pass@1 accuracy tested with greedy decoding. The results of ODPO are directly copied from [25]. Here q, r, a denote the dependency on questions, human-verified reasoning traces and golden answers respectively.

	Supervision	MATH	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
frontier model							
Llama-3.1-70B-Instruct	$\{q, r, a\}$	64.6	35.3	31.9	16.7	30.1	35.7
Eurus-2-7B-PRIME	$\{q, r, a\}$	79.2	38.6	42.1	26.7	57.8	48.9
1.5B model							
Qwen2.5-Math	None	52.2	10.7	25.2	10.0	42.5	28.1
Qwen2.5-Math-Instruct	$\{q, r, a\}$	73.8	30.9	38.7	6.7	52.5	40.5
Qwen2.5-Math w/SFT	$\{q, r, a\}$	61.8	26.1	27.1	3.3	37.5	31.2
Qwen2.5-Math w/GRPO	$\{q, a\}$	75.2	32.0	33.6	16.7	52.5	42.0
Qwen2.5-Math w/EMPO	$\{q\}$	73.0	32.4	36.6	13.3	55.0	42.1
7B model							
Qwen2.5-Math	None	64.8	15.1	26.7	6.7	40.0	30.7
Qwen2.5-Math Instruct	$\{q, r, a\}$	82.8	43.8	41.2	16.7	62.5	49.4
Qwen2.5-Math w/SFT	$\{q, r, a\}$	72.2	34.6	33.2	10.0	45.0	39.0
Qwen2.5-Math w/ODPO	$\{q, a\}$	76.8	30.9	37.9	26.7	62.5	47.0
Qwen2.5-Math w/GRPO	$\{q, a\}$	77.8	39.7	39.1	20.0	57.5	46.8
Qwen2.5-Math w/EMPO	$\{q\}$	78.0	40.4	37.3	20.0	65.0	48.1

54.94% to 70.22% on 7B model. In addition, we observe that as training progresses, the model exhibits reasoning behaviors by fully unsupervised learning. After careful considerations, the model self-corrects its previous errors and outputs the right answer. A more concrete case is shown in the appendix.

Table 2: Overall results on free-form natural reasoning benchmarks. Here w/CoT denotes Qwen2.5 Base model with the same reasoning prompt template we used to train EMPO.

	Supervision	MMLU Pro				
		STEM	Humanities	Social	Other	Avg.
frontier model						
3B model						
Qwen2.5-Base 5-shot	$\{q, r, a\}$	34.7	26.2	47.9	35.9	35.3
Qwen2.5-Instruct	$\{q, r, a\}$	44.8	30.7	56.0	47.1	44.5
Qwen2.5-Base w/EMPO	$\{q\}$	46.4	30.5	53.0	46.4	44.7
7B model						
Qwen2.5-Base	-	30.1	23.8	45.9	34.3	32.1
Qwen2.5-Base 5-shot	$\{q, r, a\}$	45.7	36.3	59.1	49.4	46.8
Qwen2.5-Instruct	$\{q, r, a\}$	56.9	38.1	64.1	58.6	55.2
Qwen2.5-Base w/EMPO	$\{q\}$	52.4	34.6	59.0	50.9	50.1

4.2.3 Training Dynamics

We further conduct experiments to investigate the reliability of our unsupervised reward signals. As shown in Figure 2, the unsupervised reward signals of EMPO have a strongly negative correlation with the true rewards based on golden answers.

5 Discussions

What Does EMPO's Strong Performance Imply? In our experiments with the 7B model, EMPO even slightly outperforms supervised methods like GRPO and Online-DPO, despite requiring no external supervision. Besides, despite the substantial improvement on common-used benchmarks, we detect no consistent increase in response length or evidence of an "Aha moment"–the hypothesized phenomenon where enhanced reasoning capability emerges through extended outputs [25]. This counterintuitive observation has led us to hypothesize potential explanations. We align our interpretation with prior insights from [32]: "*Pretraining does all the hard work. One big bet is that the pretraining phase grants all the abilities to the base LM, and finetuning is simply like a style transfer which positions*

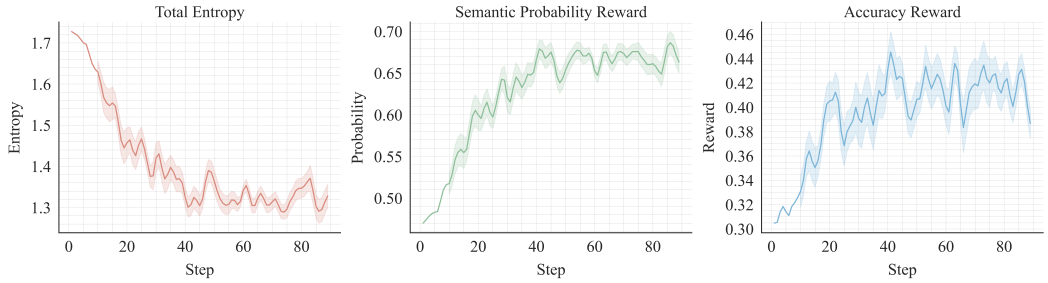


Figure 2: We visualize the training dynamics when tune Qwen2.5-Math-7B Base model with EMPO on 20K prompts randomly sampled from NuminaMath-CoT. The left illustrates the running average of semantic entropy (Eq. 4). The middle shows the trend of our unsupervised reward as defined by Eq. 6. We also visualize the model accuracy on training data at each optimization steps in the right figure. Along the unsupervised RL-based training trajectory, EMPO establishes a stable learning process with consistently decreased semantic entropy and improved accuracy.

the model to the right output space." Under this conjecture, we attribute the efficacy of our method to the robust pretraining process of the Qwen2.5-Math Base model. While the strong performance of our unsupervised approach on complex reasoning tasks was unforeseen, we believe that it could prove valuable for downstream applications where human expert annotation of gold-standard answers is cost-prohibitive and unsustainable. We are still working in progress to investigate the performance boundaries of our method and systematically evaluate the gap between unsupervised and supervised paradigms such as PPO, Online-DPO, and GRPO. In the future, we plan to further explore different LLM families (e.g., the Llama family) and training data for continuously scaling up fully-unsupervised reasoning-oriented RL.

How to Mitigate Potential Reward Hacking? Note that different from verifiable rule-based reward, which inherently resists reward hacking risks, optimizing unsupervised entropy objectives may permit trivial solutions. For instance, models could exploit the reward signal by overfitting to high-confidence predictions for the most frequent semantic clusters without carefully reasoning process. To address this, we implement a straightforward entropy thresholding strategy, restricting optimization to prompts exhibiting moderate uncertainty via dual threshold criteria, i.e., δ_{low} and δ_{high} in Eq. 7. Developing more effective unsupervised proxy objectives represents a promising direction for future research.

6 Conclusions and Future Work

In this work, we make an early attempt at incentivizing the reasoning capability of LLMs by fully unsupervised RL-based techniques. The proposed EMPO achieves competitive performance in both mathematical and free-form common-sense reasoning tasks. Our practice demonstrates that by exploring semantic entropy as an unsupervised optimization objective, the pre-train base or instruct model itself is sufficient to scale up benchmark performance on downstream reasoning tasks without any external supervision. There are several future research directions. The most essential would be exploring how to generalize EMPO to other tasks where the semantic equivalence of two model outputs is difficult to judge such as code generation. It would be non-trivial and challenging to determine whether two LLM outputs are equivalent since it may necessity the understanding of complex context. From a broader scientific perspective, it would be interesting to better understand and exploit the uncertainty of powerful LLMs.

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A Experiment Details

Prompt template. For both mathematical and natural reasoning tasks, we adopt the following reasoning prompt template similar with Online-DPO-R1 project [25] for both training and testing.

```
<|im_start|>system
Reason step by step and put the answer in \boxed{}.
<|im_end|>
<|im_start|>user
{Question} Reason step by step and put the answer in \boxed{}.
<|im_end|>
<|im_start|>assistant
```